

Basic Molecular Modelling Course for Chemists and Life Sciences I (Basic MoCo I)

Teachers (Tutors):

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Skills requirements:

Basic understanding in chemistry, Basic understanding of Biology, Ability to use a PC (Personal Computer)

System Requirements:

8GB RAM (preferably from 16GB), 250GB ROM, Good internet connectivity

Software to be Installed:

Bash terminal (WSL recommended for Windows users-----alternatives are virtual machines like Virtual Box; bash already preinstalled for MacOS users), Visual Studio Code or Jupyter Notebook for python, Chem-Sketch (or Marvin Sketch), Open Babel (as GUI or via the terminal), Discovery Studio, ChimeraX, Pymol, Modeller (Optional), VMD (optional), Vina (via bash terminal), Schrödinger

Software hosted on the Web:

Google Colab (<https://colab.research.google.com/>), Protein Data Bank (<https://www.rcsb.org/>), Uniprot Database (<https://www.uniprot.org/>), Swissadme; (<https://www.swissadme.ch/>), Swiss model (<https://swissmodel.expasy.org/>), AlphaFold (<https://alphafold.ebi.ac.uk/>), ChEMBL (<https://www.ebi.ac.uk/chembl/>), PubChem (<https://pubchem.ncbi.nlm.nih.gov/>), Binding DB (<https://www.bindingdb.org/rwd/bind/index.jsp>), Natural Products Databases (<https://coconut.naturalproducts.net/>), Drug Bank (<https://go.drugbank.com/>), Chemical Spaces (ChemDiv, Enamine, Otava)

Learning Outcomes

At the end of this course, the student should:

- Be able to understand the command line interface, structure of files and directory; to create, move and access files; to read files and extract needed information; to concatenate commands and direct outputs
- Be able to understand the basics of python API and usage for common tasks.

- Understand basic representations of chemical structures and biomolecules; how to enumerate R groups and convert between structure file types
- Learn how to navigate Uniprot and protein data bank to retrieve structures
- Learn how to perform sequence alignment, fill missing residues in proteins, carry out mutations in proteins, model unknown structures using traditional and modern tools; co-folding tools.
- Understand the concepts of virtual screening using descriptors, pharmacophores, fragments, scaffold hopping etc.
- Understand the various methods of similarity searches: Tanimoto, fingerprints etc
- Be able to understand the theory behind molecular docking; perform molecular docking using vina and Glide; validate a docking/ scoring algorithm; rank compounds based on docking score
- Be able to visualize molecular interactions in 2D and 3D; identify and interpret key interactions
- Be able to integrate all the knowledge and skills above to successfully carry out a capstone project

Format of the Course

The course is made of 9 core modules (as attached at the end of this document), each with its own teaching and practical materials.

The course is made of three essential parts:

- The teaching materials (in PDF) available for the student; to include the topic materials and the practical for each module.
- A video Recording where the tutor explained the concepts in the teaching material (Usually 30-40 minutes)
- A second video recording of the practical that follow the theory of each teaching units

Assessment

The last module of the course is a capstone project. This will not be directly related to any single idea in the different modules; rather it will be a complex real world problem which will require you to apply your understanding of many concepts and practical from the different modules.

Course Contents

Module	Description of Contents
1. Basic Linux for Life Sciences	File structures, basic commands, create and copy files, rename files, search for files, wildcards, remote connection

2.	Basic Python for Life Sciences	Basic python concepts: data types, lists, functions, load and explore scientific datasets, perform basic data cleaning, create simple visualisations, build a foundation for data science and machine learning
3.	Chemical Structures	File formats and representations (pdb mol mol2 sdf pdbqt etc), Meaning and Uses and applications, Use of chemsketch, chemdraw and other online tools for drawing and chemical enumeration, Converting between file types
4.	Targets (Proteins) selection , validation and Databases	Uniprot, The protein Data Bank, Protein retrieval and preparation, ChEMBL and ChEMBL API, PubChem and PubChem API, Binding DB for drug discovery, Coconut database, Super-Natural 3.0, Natur-Ar, Afro-Database, Drug bank, Chemical space and diversity
5.	Protein Modelling	Sequence alignment, Blast, Filling Missing residues, Swiss, Model/Modeller, AlphaFold, Co-folding
6.	Virtual Screening I	Chemical Descriptors, Descriptor Based Screening (Lipinski, Verbers etc.), Similarity and Similarity Search, Pharmacophore Screening, Fragment Based Screening
7.	Virtual Screening II	Docking Theory, Search, Scoring, Ranking, Algorithms, ML based Scoring Function, Docking Implementations, Docking Validation
8.	Virtual Screening III	Structure based virtual screening with Glide
9.	Visualization/Representations	2D & 3D visualization of molecular Interactions using Pymol, Discovery Studio and ChimeraX
10.	Capstone Project	Provides the opportunity to integrate all the knowledge and skills in all the modules. It will not be directly related to any single idea in the different modules; rather, it will be a complex real-world problem which will require you to apply your understanding of many concepts and practical skills from the different modules.

Appreciation:

To my Supervisors, Dr. Wande Oluyemi and Dr. Adeniyi Adewumi. Special thanks also to the CHPC centre in South Africa for the generous use of their computing resources